

Bharath Kumar

Lead iOS Engineer

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Profile Summary:

- **10+ years** of experience as an **iOS Developer** with a proven track record in designing, developing, and optimizing **scalable, high-performance mobile applications** for leading global clients across industries such as **E-Commerce, Banking, Pharma, Airlines, Media, Energy**, and more.
- **Successfully led the end-to-end development of an award-winning mobile application** for a Fortune 500 client, enhancing user engagement by 30% through intuitive UI/UX design and implementing advanced real-time data synchronization across devices.
- **Pioneered the integration of complex payment systems** for a major e-commerce platform, improving transaction speed and security, which resulted in a seamless checkout experience.

Key Highlights:

- Expert in developing **native iOS applications** using **Swift** and **Objective-C**, with advanced proficiency in **SwiftUI, UIKit**, integrating it with **Combine** for reactive, **Core Data, Core Animation**, and **ARKit** for building feature-rich, seamless experiences.
- Extensive experience with **cross-functional teams** to deliver end-to-end solutions using **Clean Architecture, VIPER**, and **MVVM** design patterns, ensuring **modularization, scalability**, and **maintainability** of code.
- Successfully integrated complex **payment gateways** (Apple Pay, Google Pay, Stripe) and implemented secure **in-app purchase** flows, ensuring **PCI DSS compliance** and user security.
- Strong expertise in **API integration** with **RESTful, GraphQL, WebSocket**, and **gRPC** APIs to connect iOS apps to cloud services like **AWS** and **Firebase** for real-time data syncing and efficient resource management.
- Designed and implemented **real-time messaging features** using **WebSocket**, enhancing app interactivity for **chat and social media apps** with low latency and high data throughput.
- Led the **migration of legacy apps** to modern architectures, optimizing **binary size**, improving **app performance**, and ensuring smooth user experiences in diverse environments (iPhone, iPad, Apple Watch).
- Expertise in **Xcode** project/workspace setup, modular target configuration, build schemes, and custom configurations to optimize **CI/CD and reduce build times**.
- Implemented **machine learning models** using **Core ML** and **Create ML** for use cases such as **image classification, object detection**, and **predictive modeling**, enhancing app functionality and user engagement.
- Skilled in **Xcode Cloud / Fastlane integration**, automated signing, TestFlight distribution, unit/UI test automation, and build artifact management for reliable release pipelines.
- Integrated **IoT technologies** like **BLE (Bluetooth Low Energy), HomeKit**, and **wearable devices** (Apple WatchKit, NFC) to deliver interactive, cross-platform experiences, including health and fitness apps.
- Experience with **StoreKit testing (local and sandbox)**, StoreKit configuration files, and handling off-platform edge cases (grace periods, refunds, subscription status).
- Extensive experience with **CI/CD pipelines** using **Jenkins, Fastlane, TestFlight**, and **Firebase App Distribution**, enabling **continuous delivery** and efficient version control in **Agile Scrum** environments.
- Designed and implemented **biometric authentication** (Face ID, Touch ID) for secure login, enhancing app security through **Keychain Services** and **SSL pinning** for compliance with GDPR, CCPA, and **App Transport Security (ATS)**.
- Delivered robust **data persistence strategies** with **Core Data, SQLite, Realm**, and **CloudKit**, ensuring **local storage** and **cloud syncing** of user data across devices.
- Developed advanced **AR-based apps** using **ARKit** and **SceneKit**, creating innovative experiences such as virtual product trials, augmented navigation, and educational tools for various industries.
- Spearheaded the implementation of **offline-first strategies**, enabling users to seamlessly interact with apps even without internet connectivity, syncing data once the device is online.
- Extensive knowledge of **performance optimization** techniques such as **lazy loading, memory management** (ARC, Non-ARC), **binary size reduction**, and **image asset optimization**, ensuring high-quality apps with minimal resource consumption.
- Strong experience in **user-centric design**, collaborating with design teams to create apps that are **intuitive, accessible**, and **adaptive** across multiple screen sizes and iOS versions.

- Led teams in **Agile/Scrum environments**, utilizing tools such as **Jira, Confluence, Slack**, and **Trello** to manage sprints, track progress, and ensure timely delivery of high-quality applications with minimal bugs.

Technical Skills:

Programming Languages & Frameworks

- **Languages:** Swift, Objective-C, Kotlin, Java, C, C++, XML, Python, Rust
- **Frameworks:** SwiftUI, UIKit, Core Data, Combine, Core Graphics, Core Animation, Cocoa Touch, RxSwift, ReactiveCocoa, CoreBluetooth, ARKit, SceneKit
- **Design Patterns:** MVC, MVVM, VIPER
- **Design & Architecture:** Clean Architecture, Dependency Injection, Modularization

Development Tools

- **IDE & Build Tools:** Xcode, Interface Builder, CocoaPods, Carthage, Swift Package Manager (SPM), LLDB, Instruments (Time Profiler, Allocations, Leaks), Bazel, Gradle
- **Performance Optimization Techniques:** Memory Management (ARC & Non-ARC), Binary Size Reduction, Image Asset Optimization, Lazy Loading
- **Testing & Debugging:** XCTest, UIAutomation, OCMock, Charles Proxy, Bugzilla, Postman, CircleCI, GitHub Actions, Azure Pipelines, Unit Testing, UI Testing

Integration & Databases

- **API Integration:** RESTful APIs, GraphQL, WebSocket, Firebase, gRPC, Protobuf, Amazon DynamoDB, Realm, Redis
- **Databases:** Core Data, SQLite, MySQL, PostgreSQL, NoSQL Databases (Cassandra, Firebase Realtime Database)
- **Cloud Services:** AWS (S3, Lambda, Cognito), Azure

CI/CD & Version Control

- **Version Control:** Git, GitHub, GitLab, Bitbucket, Subversion (SVN)
- **CI/CD Tools:** Jenkins, Fastlane, TestFlight, Firebase App Distribution

Security & Compliance

- **Technologies:** Biometric Authentication (Face ID, Touch ID), Secure Enclaves, Keychain Services
- **Compliance:** SSL Pinning, OAuth 2.0, Data Encryption, Memory Management (ARC & Non-ARC), GDPR, CCPA, App Transport Security (ATS), Certificate Pinning

Machine Learning & AI

- **Technologies:** Core ML, Create ML, TensorFlow Lite, ONNX, NLP
- **Use Cases:** Image Classification, Object Detection, Natural Language Processing, Predictive Modeling

IoT & Hardware Integration

- **IoT Technologies:** BLE (Bluetooth Low Energy), NFC, Zigbee, MQTT, LiDAR, BLE, NFC, HomeKit, Custom Hardware Integration
- **Hardware Integration:** Wearable Devices, GPS, Accelerometer, Gyroscope, Apple WatchKit, iBeacon, Sensors

Payments & E-Commerce

- **Payment Integration:** Apple Pay, Google Pay, Stripe, PayPal, Square APIs, Plaid API, Payment Gateway Customization, PCI DSS Compliance
- **E-Commerce Features:** In-App Purchases, Secure Transactions, Subscription Management

Other Technical Expertise

- **Cross-Platform:** React Native, Flutter, Kotlin Multiplatform (KMM)
- **Agile Tools:** Jira, Confluence, Trello, Slack

Professional Experience:

Lead iOS Developer | American Airlines, Fort Worth, TX | April 2025 - Present

iOS App Link: <https://apps.apple.com/us/app/american-airlines/id382698565>

Project Summary: I contributed to the American Airlines iOS app by enhancing key modules including AAdvantage, Promotions, Secure Traveler Info, and Travel Credits. Led the modernization of legacy MVC components to MVVM, improving scalability, testability, and performance. Integrated GraphQL for efficient data handling, optimized lazy loading for smoother user experiences, and ensured full accessibility compliance aligned with AA's design system.

Responsibilities:

- Led development of the American Airlines iOS app, overseeing feature enhancements and architectural modernization within a SwiftUI and UIKit hybrid codebase to ensure consistency, performance, and maintainability.
- Designed and implemented MVVM architecture with modular frameworks such as *AmericanCoreUI* and *AmericanBusinessUI*, enabling scalable development and code reuse across multiple modules.
- Integrated GraphQL for efficient data fetching with REST API fallbacks, improving network performance and reducing payload size for high-traffic modules like *AAdvantage* and *Travel Credits*.
- Built Combine-based data streams for real-time updates in the *AAdvantage* and *Travel Credits* modules, enhancing responsiveness and user engagement.
- Delivered high-impact features including Secure Traveler forms, Promotions carousel, and Travel Credit cards, ensuring seamless user interaction and backend synchronization.
- Implemented dependency injection to decouple components, enhance testability, and improve overall maintainability of the app's codebase.
- Championed unit and snapshot testing using XCTest, ensuring reliable and regression-free releases across all major modules.
- Streamlined release management by automating build, test, and deployment pipelines using Fastlane and Xcode Cloud, reducing manual intervention and accelerating delivery cycles.
- Drove major accessibility improvements by implementing VoiceOver support, Dynamic Type, and color contrast adjustments, ensuring compliance with WCAG and AA's accessibility design standards.
- Collaborated cross-functionally with backend, QA, design, and product teams to deliver high-quality, user-centric features aligned with business objectives.

Lead iOS Engineer | Home Depot, Atlanta, GA | February 2022 – March 2025

iOS App Link: <https://apps.apple.com/us/app/the-home-depot/id342527639>

Project Summary: I contributed to the development and optimization of The Home Depot iOS app, focusing on performance, new features, and UI enhancements. I worked on implementing voice search for hands-free shopping, personal recommendations, and streamlined checkout processes. I integrated advanced payment systems like Apple Pay for secure transactions and collaborated with backend teams to ensure seamless real-time data synchronization.

Responsibilities:

- Developed and optimized the shopping, search, and checkout functionalities for an improved user experience using SwiftUI and UIKit.
- Leveraged SwiftUI's declarative syntax to build dynamic and reusable UI components, reducing development time and improving code maintainability.
- Implemented a streamlined checkout process, optimizing UI elements for faster transactions.
- Developed Core Data models to manage offline data and sync user data with the cloud using Firebase.
- Integrated RESTful APIs and GraphQL to fetch product information, reviews, and customer data in real time.
- Utilized RxSwift and Combine for reactive programming, handling asynchronous tasks such as data fetching and UI updates.
- Ensured data security by integrating OAuth 2.0, SSL Pinning, and Biometric Authentication using Face ID and Touch ID.
- Optimized memory management using ARC, lazy loading, and reducing binary size to improve app performance.
- Implemented in-app purchases and managed Apple Pay and Google Pay integration for secure payment transactions.
- Integrated NFC technology for enhanced interactions with physical products.
- Worked on backend integration with AWS Lambda for processing user data and updating product catalogs in real time.
- Managed CI/CD pipelines using Jenkins and automated app deployment with Fastlane and TestFlight.
- Worked in an Agile Scrum environment with regular stand-ups, sprints, and retrospectives using Jira and Confluence.
- Conducted performance testing using Instruments to measure memory usage, CPU performance, and network optimization.
- Contributed to code reviews and maintained code quality by following clean code principles and using Swift Package Manager (SPM) for dependency management.
- Collaborated with cross-functional teams, including backend engineers and product managers, to define new features and resolve issues.
- Tracked user analytics using Google Analytics and Firebase Analytics to improve user engagement.
- Implemented the MVVM (Model-View-ViewModel) architecture to separate business logic from UI components, enhancing code readability, testability, and maintainability in iOS applications.

Lead iOS Developer | Truist Bank, Charlotte, NC | December 2020 - January 2022

iOS App Link: <https://apps.apple.com/us/app/truist-mobile/id1555389200>

Project Summary: I worked on enhancing Truist Bank's iOS mobile banking experience, focusing on improving app performance, security features, and user engagement. I integrated advanced transaction monitoring features, optimized real-time account notifications, and added biometric authentication for secure logins. I collaborated with the backend team to ensure secure data handling and built features like mobile check deposits and loan applications.

Responsibilities:

- Developed features for mobile check deposit, enabling users to securely capture and upload check images for processing.
- Used Swift concurrency (async/await) for safe multi-threading and XCTest with Slather for coverage reporting
- Integrated Face ID and Touch ID to provide secure login and transaction verification for users.
- Enhanced real-time notifications for account balances, payments, and transfers using APNs (Apple Push Notification service).
- Optimized the account overview page to provide users with a seamless, personalized view of their financial status.
- Integrated RESTful APIs for transaction data and real-time updates from backend services.
- Utilized Core Data to handle offline storage and sync data across devices while maintaining data consistency.
- Built custom UI components using SwiftUI and UIKit to deliver a smooth and interactive user experience.
- Employed Core Data and SQLite for efficient data management and local storage, utilizing Combine to streamline asynchronous data handling and improve performance.
- Implemented OAuth 2.0 authentication to securely manage user credentials and protect sensitive data.
- Leveraged RxSwift to handle asynchronous events, data streams, and UI updates efficiently.
- Integrated Plaid API for seamless bank account linking and real-time transaction monitoring.
- Conducted performance tuning using Instruments for memory, CPU usage, and data load optimization.
- Developed transaction history screens using Core Animation to provide smooth data transitions.
- Optimized app loading speed and reduced binary size to improve app performance and reduce download time.
- Implemented security features like SSL Pinning, data encryption, and certificate pinning to prevent man-in-the-middle attacks.
- Collaborated with backend teams to integrate AWS Lambda for processing financial transactions and account updates.
- Worked with the CI/CD pipeline using Jenkins and Fastlane to automate the testing and deployment process.
- Contributed to team discussions and led sprint retrospectives within the Agile Scrum framework using Jira and Confluence.
- Worked with SQLite for local data storage and syncing across user devices.
- Implemented the MVVM (Model-View-ViewModel) architecture to streamline data flow and improve separation of concerns in iOS applications, ensuring scalable and maintainable code.
- Developed cross-platform mobile applications using React Native, leveraging its framework to build reusable components for both iOS and Android platforms.
- Integrated native iOS modules with React Native to enhance app performance and access platform-specific features, ensuring a seamless user experience.

Senior iOS Developer | Amgen Inc., CA | August 2019 - November 2020

iOS App Link: <https://apps.apple.com/ma/app/amgen-events/id1210873799>

Project Summary: I worked on Amgen Inc.'s iOS mobile app to enhance user experience and provide advanced functionalities like medication tracking, health insights, and direct communication with healthcare providers. My primary focus was on integrating machine learning models for personalized health recommendations and enhancing the overall app performance. I implemented secure login flow using biometrics and integrated real-time updates through push notifications to keep users informed.

Responsibilities:

- Contributed to the development of a personalized medication tracker and health insights feature.

- Developed a real-time notification system for alerts on prescription renewals and health tips.
- Integrated Core ML for personalized health recommendations based on user data.
- Developed a biometric authentication flow (Face ID/Touch ID) to enhance security for sensitive medical data.
- Implemented background sync functionality for real-time updates and seamless offline access.
- Integrated Firebase for data storage and push notifications.
- Optimized app performance, reducing loading time by 40% using image asset optimization.
- Implemented in-app purchase feature for premium healthcare services and subscription management.
- Integrated real-time communication features using WebSocket and Combine, optimizing push notifications and background data syncing for a seamless user experience.
- Used RxSwift for handling asynchronous tasks and improving the code's reactivity.
- Led debugging and troubleshooting efforts to ensure high-quality standards across all app features.
- Wrote comprehensive unit and UI tests using XCTest and UIAutomation.
- Designed custom user interfaces with SwiftUI for better user engagement.
- Ensured GDPR and HIPAA compliance by implementing strong data encryption and secure storage.
- Automated app deployment using Fastlane and integrated CI/CD pipelines with Jenkins.
- Leveraged Git for version control and collaborated in an Agile environment with Jira.
- Used Xcode Instruments for memory management and performance profiling.
- Integrated Apple HealthKit to allow users to sync health data with the app.
- Developed and optimized RESTful API integrations for user health data.
- Enhanced app scalability by modularizing the code and applying Clean Architecture principles.

Senior iOS Developer | Hawaiian Airlines, Honolulu, HI (Remote) | June 2018 - July 2019

iOS App Link: <https://apps.apple.com/us/app/hawaiian-airlines/id575400964>

Project Summary: I worked on the Hawaiian Airlines mobile app to enhance the in-flight experience, improve flight tracking, and provide real-time notifications for travelers. I developed features such as personalized travel itineraries, seat selection, and seamless in-app payment integration. Additionally, I focused on enhancing the app's user interface and performance to ensure a smooth experience even in low connectivity conditions.

Responsibilities:

- Developed a personalized itinerary feature to show passengers their upcoming flights and relevant travel details.
- Implemented a seat selection feature with dynamic seat maps using Core Graphics and UIKit.
- Integrated in-app purchase and payment features for flight upgrades and seat selections.
- Integrated real-time flight status updates, including delays, cancellations, and gate changes using RESTful APIs.
- Designed and developed a real-time push notification system to keep users updated on their flights.
- Optimized the app for offline functionality by storing critical data locally using Core Data.
- Used Firebase for user authentication and real-time push notifications.
- Integrated Apple Wallet to allow users to store boarding passes for easy access.
- Enhanced app performance by optimizing image assets and reducing app size by 30%.
- Led debugging and troubleshooting efforts to resolve critical issues in production.
- Automated deployment processes using Fastlane and integrated CI/CD workflows with Jenkins.
- Designed and implemented a custom, modular codebase using MVVM architecture for scalability.
- Ensured high app security standards, implementing SSL pinning and data encryption.
- Applied Agile methodologies and worked closely with product managers using Jira and Confluence.
- Used XCTest for unit and UI testing to ensure functionality across all app features.
- Integrated Apple Pay for secure in-app payments and flight upgrades.
- Developed dynamic user interfaces with SwiftUI and UIKit to improve user experience.
- Used Xcode Instruments for performance profiling and memory leak detection.
- Developed API integrations with internal services to manage bookings and check-in.

iOS Developer | Hulu, Santa Monica, CA | March 2017 - May 2018

iOS App Link: <https://apps.apple.com/us/app/hulu-stream-tv-shows-movies/id376510438>

Project Summary: I contributed to the Hulu: Stream TV shows & movies app, focusing on enhancing content streaming and providing personalized recommendations based on users' viewing history. I integrated a new video player feature with support for adaptive streaming and implemented a seamless login experience through Single Sign-On (SSO) to provide users with a consistent experience across multiple platforms. My role was key to improving user engagement and accessibility for live and on-demand content.

Responsibilities:

- Developed and enhanced the in-app video player to support adaptive streaming for live and on-demand video content.

- Integrated Single Sign-On (SSO) authentication for seamless login across multiple Hulu Network platforms.
- Implemented content recommendations based on machine learning algorithms and user behavior analytics.
- Enhanced in-app video player using AVFoundation and HLS; implemented adaptive bitrate with VideoToolbox where applicable
- Optimized content caching for faster video loading and reduced buffering using Core Data and custom caching mechanisms.
- Integrated analytics via Mixpanel and Amplitude; used Sentry/Crashlytics for crash reporting
- Worked on a seamless multi-device streaming experience allowing users to start watching on one device and resume on another.
- Integrated real-time video playback analytics to monitor user engagement and improve recommendations.
- Developed and maintained push notification features for live events and personalized alerts for new content.
- Created custom UI components for video controls, playlists, and content navigation using UIKit and Core Animation.
- Integrated third-party libraries for video streaming protocols like HLS (HTTP Live Streaming) and DASH (Dynamic Adaptive Streaming over HTTP).
- Ensured secure video playback by implementing SSL Pinning and encrypted video streams.
- Integrated external ad servers and implemented in-app advertising and monetization features (pre-roll, mid-roll, and post-roll ads).
- Optimized the app for background tasks, ensuring smooth playback during background operation.
- Developed features that allowed users to save content to their watchlist and manage their subscriptions.
- Worked on accessibility features like closed captions, subtitle options, and support for VoiceOver.
- Integrated Firebase for real-time analytics, crash reporting, and performance monitoring.
- Implemented data persistence using Realm to store user preferences and content metadata.
- Worked on app performance optimization using Instruments, reducing memory usage and improving app load time.

iOS Developer | Holland America Line Inc, Seattle, WA | January 2016 - February 2017

iOS App Link: <https://apps.apple.com/us/app/holland-america-line-navigator/id1450557920>

Project Summary: I led the development of a feature-rich iOS app that enhances the passenger experience for Holland America Line guests. The app provides real-time itinerary tracking, interactive maps, onboard booking management, and seamless communication with crew members. I integrated geolocation features to track the ship's location in real-time and developed a personalized notifications system to keep passengers informed about key events, activities, and dining options.

Responsibilities:

- Developed a seamless user interface for viewing cruise itineraries and events using UIKit.
- Implemented interactive ship maps, allowing users to view the ship's real-time location and nearby ports of call.
- Integrated a secure booking system, enabling passengers to manage dining reservations, shore excursions, and other onboard activities.
- Built a push notification system that sends personalized alerts for important events, such as special promotions and upcoming excursions.
- Designed and implemented a secure onboard messaging system for passengers to communicate with the crew through WebSocket and Firebase.
- Worked with CoreBluetooth to integrate wearable smartband functionality, allowing passengers to access ship amenities with ease.
- Leveraged RESTful APIs to fetch and update real-time data on ship schedules, available activities, and dining options.
- Optimized app performance using advanced memory management techniques, ensuring smooth performance during high-traffic periods.
- Utilized MapKit to display ship location, port information, and itineraries, providing users with a navigational tool throughout their cruise.
- Implemented CoreData to store user preferences and offline content, allowing the app to function without internet connectivity.
- Led the integration of third-party payment systems to allow users to make in-app purchases for onboard services.
- Worked with the backend team to synchronize passenger data and bookings with the server, ensuring accuracy and real-time updates.
- Ensured app security by implementing SSL pinning, OAuth 2.0, and GDPR compliance measures.

- Collaborated with cross-functional teams to align app development with business goals and user needs.
- Led the integration of analytics to track app usage, engagement, and user behavior, providing insights for future updates.
- Used Fastlane to automate build and deployment processes, streamlining the CI/CD pipeline.
- Conducted unit and UI testing to ensure app stability and performance across different iOS devices.

iOS Developer | Atlassian, San Francisco, CA | January 2015 - December 2015

iOS App Link: <https://apps.apple.com/us/app/confluence-data-center/id1288365159>

Project Summary: I led the development of a high-performance enterprise iOS app Confluence Data Center, providing a seamless platform for collaboration, communication, and real-time data analysis. The app integrates advanced machine learning models for predictive insights, securely synchronizes data across devices, and facilitates smooth collaboration.

Responsibilities:

- Developed and optimized the real-time navigation system to deliver accurate, up-to-the-minute traffic information based on crowdsourced data.
- Implemented geolocation-based features to provide turn-by-turn navigation and real-time route planning for users.
- Designed the user interface to ensure ease of use while displaying complex traffic data such as accidents, road closures, and speed limits.
- Utilized Swift to build scalable and maintainable iOS app features, enhancing team communication, and task management with UIKit and CoreData.
- Implemented real-time data synchronization with Firebase for messaging and task updates, ensuring seamless collaboration among team members.
- Integrated CoreML for intelligent task predictions, leveraging machine learning to analyze employee behavior and recommend productivity-enhancing actions.
- Enabled biometric login with Touch ID and Face ID using Swift to provide secure, quick access to the app.
- Developed custom views and animations using Core Animation and Swift, enhancing the visual experience and engagement within the app.
- Integrated Google APIs (Maps, Drive, Firebase) to enable features like real-time location tracking, document management, and seamless data syncing.
- Implemented location-based services using MapKit and CoreLocation to provide real-time office navigation and task assignments based on proximity.
- Leveraged RESTful APIs to fetch data from internal Atlassian services and ensure real-time updates within the app.
- Built a real-time collaborative document editing feature using Google APIs, allowing multiple users to edit documents simultaneously.
- Optimized app performance through detailed profiling using Instruments and Time Profiler, ensuring high responsiveness and fast load times.
- Implemented offline functionality, using CoreData to store key data and ensure users can access and modify information without an internet connection.
- Led the integration of secure network communications using OAuth 2.0 for authentication and SSL/TLS encryption for data protection.
- Enhanced Bluetooth-based communication with CoreBluetooth, supporting seamless connectivity with IoT devices in the workplace.
- Automated testing and CI/CD pipelines using XCTest, UIAutomation, and Fastlane for efficient development cycles and improved deployment.

Education:

Bachelor's in computer science
Master's in computer science
 Kennesaw State University